

B3-BCC-STRUCT refutation and retirement (operator verdict 2026-06-23)

TECT verification-first repository · claim B3-BCC-STRUCT

first issued 2026-06-23 · this version issued 2026-06-23 · v1.1

1. Purpose and scope

This note records the refutation of B3-BCC-STRUCT and its retirement per the operator verdict of 2026-06-23. It makes no positive claim. The refutation evidence (§2) is accepted; the v1.0 proposal to *reframe* B3 onto B1 is rejected (§3) and B3 is retired (§4).

2. Refutation evidence (accepted)

The seeded B3 evidence is the single-shell SMA structural ranking (Math194) and the 1-mode K_4/K_6 ranking (Math383). Re-running [archive/legacy/scripts/Math194_brazovskii_lattice_ranking.py](#) places **BCC at rank 9 of 10** (lamellar rank 1, $F/V = -1.211 \times 10^{-5}$; bcc $F/V = -4.845 \times 10^{-6}$) — the opposite of $F_{\text{BCC}} < F_{\text{FCC}} < F_{\text{SC}}$. Independently, **Math400** (2026-05-11, T0 binding) shows that at $\mu^2 = +0.005$ every lattice collapses to the disordered $F = 0$ state (the SMA “BCC minimum” is a saddle) and refutes the Math383 K -table. Registered: [negative-results/registry.md](#) R-2026-06-23-b3-bcc-structural-selection.

3. Why the reframe is rejected (operator verdict)

B1-RH-ENUM establishes the *restricted* proposition

$$\Delta F_{\text{enum}}[\mathcal{R}] := F_{\text{TECT}}[\mathcal{R}] - F_{\text{TECT}}[\mathcal{R}_H] > 0, \quad \mathcal{R} \in \mathcal{E}_{\text{tested}}, \quad (1)$$

a *relative ranking within the tested ordered-reading ensemble*. It does **not** entail

$$F[\mathcal{R}_H] < F[0] \quad \text{nor} \quad H_{\mathcal{R}_H} \succeq 0 \text{ under unrestricted variations,} \quad (2)$$

and must not be conflated with the Math400 disordered-collapse / saddle result. Therefore B1 cannot resurrect the original “BCC energy-condensate structure is selected” claim. Adversarial points upheld by the operator: (i) “continuum- anchored” must mean a continuum-functional *estimator*, not a completed $a \rightarrow 0$ PDE continuum limit (PDE continuum certification is open); (ii) keeping the original card name for a meaning-replaced claim is a meaning substitution, not a scope reduction; (iii) the reframed ranking falsifier is not a falsifier of any BCC phase/minimum claim.

4. Disposition (retirement)

B3-BCC-STRUCT $\rightarrow$ **T0 REFUTED / RETIRED**. The surviving restricted result is carried by the separate, B1-dependent card B3-RH-TESTED-STRUCTURE-RANKING (T4, estimator grade): $\mathcal{R}_H$ is selected within $\mathcal{E}_{\text{tested}}$. B1-RH-ENUM’s own T7 scope is unaffected. Physical BCC condensate existence/stability/global selection is **not established** by any current card.

5. Devil’s-advocate (three objections)

- (α) “*The restricted ranking should keep the B3 name.*” DISMISSED: the operator ruled that reusing the name after the physical claim is refuted is meaning substitution; the restricted result gets a new, explicitly projection-named card.
- (β) “*Math194’s rank-9 might be a buggy script.*” DISMISSED on the refutation direction: the script’s *ranking* (lamellar lowest, BCC near-highest) is corroborated independently by Math400’s symmetry-preserving multi-shell solver; only the script’s self-verdict logic (“confirmed”) was buggy, not the ordering.
- (γ) “*Retiring B3 breaks C6.*” VALID with mitigation: C6’s B3-BCC-STRUCT dependency was re-pointed to B3-RH-TESTED-STRUCTURE-RANKING with an operator-review flag, since C6’s BCC-structure premise now rests only on the restricted ranking, not on an established physical selection.

Quantitative sanity check (mandatory): the Math194 re-run ratios are internally consistent ($C_{\text{lat}}^{\text{bcc}} = 3.75$, $F/V \propto -A_{\text{min}}^2$ reproduces -4.845×10^{-6}); lamellar $C_{\text{lat}} = 1.5$ (Math400-corrected) is the deepest, confirming the ranking is the genuine single-shell SMA result — which is exactly why that reduction is refuted in favour of a continuum-anchored *estimator*.

6. Result footer

Result ID:	B3-BCC-STRUCT (REFUTED / RETIRED)
Precise statement:	The fixed-ordered BCC structural-selection claim F_BCC<F_FCC<F_SC is refuted (Math194 rank 9; Math400 disordered collapse/saddle). The reframe onto B1 is REJECTED (operator 2026-06-23). B3 retired.
Scope:	No positive claim. Surviving restricted projection: B3-RH-TESTED-STRUCTURE-RANKING (T4).
Dependencies:	negative-results R-2026-06-23-b3-bcc-structural-selection.
Evidence grade:	EXECUTED (refutation demo).
Reproduction command:	python archive/legacy/scripts/ Math194_brazovskii_lattice_ranking.py (-> BCC rank 9)
Falsification gate:	n/a (refuted).
Tier before / after:	B3 T4 / T0 REFUTED.
No-overclaim statement:	No BCC phase/minimum/global selection; B1’s relative ranking is NOT a physical-selection claim.
Next required action:	Re-establish BCC structure as a NEW claim by certifying F[Psi_min]<F[0], lambda_min^perp>=0 (symmetry- projected), N->inf on the canonical PDE background.